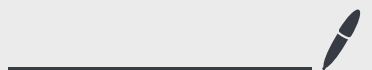
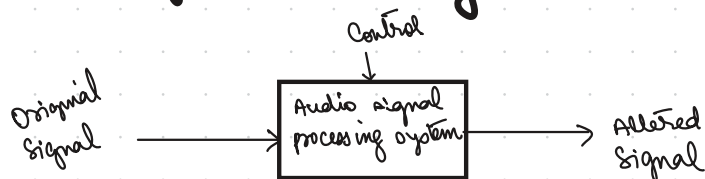


- 1) Discrete Fourier Transform
- 2) Fourier Transform Properties
- 3) Short Time Fourier Transform
- 4) Sinusoidal Model
- 5) Harmonic Model
- 6) Sinusoidal plus residual modelling
- 7) Sound Transformations
- 8) Sound/Music description



Audio Signal Processing



Complex sine wave

$$\begin{aligned}\bar{x}[n] &= A e^{j(\omega n T + \phi)} = A e^{j\phi} e^{j\omega n T} = x e^{j(\omega n T)} \\ &= A \cos(\omega n T + \phi) + j A \sin(\omega n T + \phi)\end{aligned}$$

Real sine wave

$$\begin{aligned}x[n] &= A \cos(\omega n T + \phi) = A \left(\frac{e^{j(\omega n T + \phi)} + e^{j(\omega n T + \phi)}}{2} \right) \\ &= \frac{1}{2} x e^{j(\omega n T)} + \frac{1}{2} x^* e^{j(\omega n T)} = \frac{1}{2} \bar{x}[n] + \frac{1}{2} \bar{x}^*[n] \\ &= \mathcal{R} \{ \bar{x}[n] \}\end{aligned}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{j2\pi k n / N} \quad k \in [0: N-1]$$

n : discrete time index

k : discrete frequency index

ω_k : $2\pi k / N$: frequency in radians/second

f_k : $\delta_s k / N$ (δ_s : sampling rate)

Complex Exponentials

$$s_k^* = e^{j2\pi k n / N} = \cos(2\pi k n / N) - j \sin(2\pi k n / N)$$

for $N=4$ thus for $n=0, 1, 2, 3$ $k=0, 1, 2, 3$

Scalar Product

$$\langle x, s_k \rangle = \sum_{n=0}^{N-1} x[n] s_k^*[n] = \sum_{n=0}^{N-1} x[n] e^{j2\pi kn/N}$$

Tells which part of the frequency is present in the signal

DFT of complex sinusoid

$$x_1[n] = e^{j2\pi k_0 n/N} \quad \forall \quad n \in 0:N-1$$

$$\begin{aligned} X_1[k] &= \sum_{n=0}^{N-1} x_1[n] e^{j2\pi kn/N} \\ &= \sum_{n=0}^{N-1} e^{j2\pi k_0 n/N} e^{j2\pi kn/N} \\ &= \sum_{n=0}^{N-1} e^{j2\pi (k-k_0) n/N} \\ &= \frac{1 - e^{j2\pi (k-k_0)}}{1 - e^{j2\pi (k-k_0)/N}} \quad (\text{sum of geometric series}) \end{aligned}$$

if $k \neq k_0$, denominator $\neq 0$ and numerator $= 0$

thus $X_1[k] = N$ for $k = k_0$ and $X_1[k] = 0$ for $k \neq k_0$

$$\text{when } k \neq k_0, \quad \frac{e^{j2\pi (k-k_0)} - 1}{1 - e^{j2\pi (k-k_0)/N}} \quad k - k_0 \in \mathbb{Z}$$

$$= 1 - (\cos(2\pi m) + j \sin(2\pi m))$$

$$= 1 - 1 = 0$$

$$X_1[k] = 0$$

Denominator not equal to zero till $(k-k_0)$ not a multiple of N .

for $k = k_0$

$$X_1[k] = \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}k_0 n} e^{j\frac{2\pi}{N}k n} = \sum_{n=0}^{N-1} e^{j2\pi(k-k_0)n/N}$$

$$X_1[k_0] = \sum_{n=0}^{N-1} e^0 = \sum_{n=0}^{N-1} 1 \quad (1)$$

$$X_1[k] = N$$

DFT of complex sinusoid

$$x_2[n] = e^{j(2\pi f_0 n + \varphi)} \quad \forall n \in 0:N-1$$

$$\begin{aligned} X_2[k] &= e^{j\varphi} \sum_{n=0}^{N-1} e^{j2\pi(k/N - f_0)n} \\ &= e^{j\varphi} \left(\frac{1 - e^{j2\pi(k/N - f_0)N}}{1 - e^{j2\pi(k/N - f_0)}} \right) \end{aligned}$$

DFT of real sinusoids

$$x_3[n] = A_0 \cos(2\pi f_0 n/N) = \frac{A_0}{2} e^{j2\pi k_0 n/N} + \frac{A_0}{2} e^{-j2\pi k_0 n/N}$$

$$X_3[k] = \sum_{n=-N/2}^{N/2-1} x_3[n] e^{j2\pi k n/N}$$

$$= \sum_{n=-N/2}^{N/2-1} \left(\frac{A_0}{2} e^{j2\pi k_0 n/N} + \frac{A_0}{2} e^{-j2\pi k_0 n/N} \right) e^{j2\pi k n/N}$$

$$= \sum_{n=-N/2}^{N/2-1} \frac{A_0}{2} e^{j2\pi k_0 n/N} e^{j2\pi k n/N} + \sum_{n=-N/2}^{N/2-1} \frac{A_0}{2} e^{-j2\pi k_0 n/N} e^{j2\pi k n/N}$$

$$= \sum_{n=-N/2}^{N/2-1} \left(\frac{A_0}{2} \right) e^{-j2\pi(k-k_0)n/N} + \sum_{n=-N/2}^{N/2-1} \left(\frac{A_0}{2} \right) e^{j2\pi(k+k_0)n/N}$$

$$= N \begin{pmatrix} A_0 \\ 2 \end{pmatrix} \quad \text{for } k=k_0, -k_0, \quad 0 \text{ for rest of } k$$

Inverse DFT

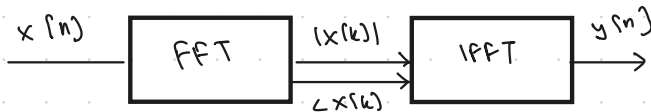
$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N} \quad x[n] \in 0:N-1$$

Inverse DFT for real signals

$$X[k] = |X[k]| e^{j\angle X[k]} \quad \text{and} \quad X[-k] = |X[k]| e^{-j\angle X[k]}$$

$$\text{for } k=0, 1, 2, \dots, \frac{N}{2}-1$$

DFT Model present in SMS Tools



Short Time Fourier Transform

We apply FFT on small segments of the signal.

Window Size: No. of samples that we apply windowing to

Frame Size:- No. of samples that are considered in each chunk of the signal after segmentation that we pass to the STFT model

$$\text{Window Size} \leq \text{Frame Size}$$

When window size < frame size, windowing function is applied to the window size and the difference b/w frame and window size is zero padded.

Algorithm :-

- (a) Take a window size, frame size, apply FFT
- (b) Go to the next window, frame size and apply FFT
- (c) Keep doing until the end of the signal.

In case of overlapping frames

Group Size (K) is important

FROM DFT TO STFT

$$\hat{x}(k) = \sum_{n=0}^{N-1} x[n] e^{j2\pi n \frac{k}{N}}$$

$$S(m, k) = \sum_{n=0}^{N-1} x(n + mN) \cdot w(n) \cdot e^{j2\pi n \frac{k}{N}}$$

frame frequency $n=0$ $\rightarrow$ starting sample of the current frame

Outputs

Spectral Matrix (# freq bins, # frames)
Complex Fourier Coefficients

$$\# \text{ frequency bins} = \frac{\text{frame size}}{2} + 1$$

$$\# \text{ frames} = \frac{\text{samples} - \text{frame size}}{\text{hop size}} + 1$$

Time/Frequency Trade off

Diagram illustrating the relationship between frame size, frequency resolution, and time resolution:

- Frame size (red arrow pointing up) is inversely proportional to Frequency resolution (green arrow pointing down).
- Frequency resolution (green arrow pointing down) is inversely proportional to Time resolution (green arrow pointing up).

Example: Transform of a windowed sine-wave (Effect of windowing)

$$x[n] = A_0 \cos(2\pi k_0 n / N) = \frac{A_0}{2} e^{j2\pi k_0 n / N} + \frac{A_0}{2} e^{-j2\pi k_0 n / N}$$

$$X[k] = \sum_{n=-N/2}^{N/2-1} w[n] x[n] e^{-j2\pi k n / N}$$

$$= \sum_{n=-N/2}^{N/2-1} w[n] \left(\frac{A_0}{2} e^{j2\pi k_0 n / N} + \frac{A_0}{2} e^{-j2\pi k_0 n / N} \right) e^{-j2\pi k n / N}$$

$$= \frac{A_0}{2} \left(\sum_{n=-N/2}^{N/2-1} w[n] e^{-j2\pi (k-k_0) n / N} + \sum_{n=-N/2}^{N/2-1} w[n] e^{-j2\pi (k+k_0) n / N} \right)$$

$$X[k] = \frac{A_0}{2} W[k-k_0] + \frac{A_0}{2} W[k+k_0]$$

Windows

Rectangular Window

(Main-lobe width = 2 bins
side-lobe level = -13.3 dB)

$$w[n] = 1 \quad n = -M/2, \dots, 0, \dots, M/2$$

$$= 0 \quad \text{elsewhere,}$$

$$W[k] = \frac{\sin(\pi k)}{\sin(\pi k / M)}$$

Hamming Window (Main-lobe width: 4 bins
side-lobe level: -31.5 dB)

$$w = 0.5 + 0.5 \cos(2\pi n / M)$$

$$n = -M/2, \dots, 0, \dots, M/2$$

$$W[k] = 0.5 D[k] + 0.25 (D[k-1] + D[k+1])$$

$$\text{where } D[k] = \frac{\sin(\pi k)}{\sin(\pi k / M)}$$

Hanning Window

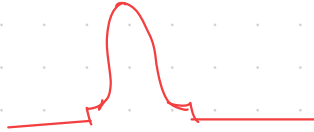
(Main-lobe width: 4 bins
Side-lobe level: -42.7 dB)

$$w(n) = 0.54 + 0.46 \cos\left(\frac{2\pi n}{M}\right) \quad n = -M/2, \dots, 0, \dots, M/2$$

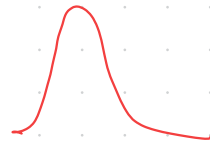
Rectangular
Window



Hanning
window

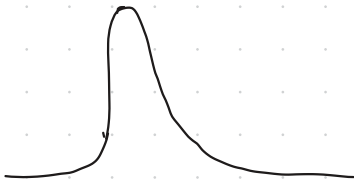


Hanning
window



Blackman Window

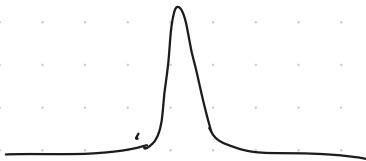
(Main lobe width: 6 bins
Side lobe level: -58 dB)



Blackman-Tukey window

(Main lobe width: 8 bins
side-lobe level: -92 dB)

(Non-existent side lobes)



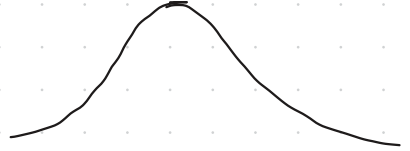
Even - Odd Size Window

Magnitude remains same, phase changes

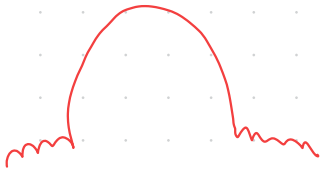
$M=32, w1$



$M=31, w2$



Magnitude:- $mW1$



$mW2$



Phase:- $pW1$



$pW2$



Hop Size

No. of samples the window slides forward before calculating the next FFT frame.

Most windows (Hamming, Hann) taper to zero at the edges.

We must overlap these windows to avoid amplitude loss (modulation).

at the frame boundaries.

We can use rectangular windows but it would lead to spectral leakage.

Smaller hop-size - higher time resolution, requires high computational power

larger hop-size :- lower time-resolution, faster processing.

Inverse STFT

$$y[n] = \sum_{l=0}^{L-1} \text{shift}_{lH, n} \left(\frac{1}{N} \sum_{k=-\frac{N}{2}}^{\frac{N}{2}-1} X_l[k] e^{j2\pi kn/N} \right)$$

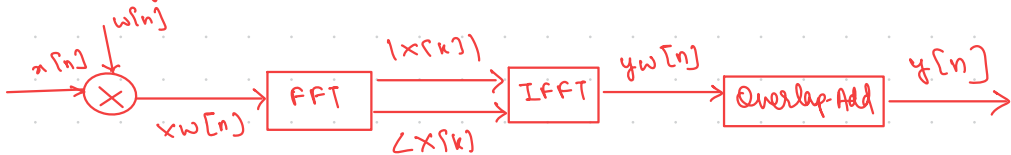
Each of p frame

$$y_{w_l}[n] = x(n + lH) w[n]$$

and the ofp sound is

$$y[n] = \sum_{l=0}^{L-1} y_{w_l}[n] = x[n] \sum_{l=0}^{L-1} w[n - lH]$$

STFT system



Sinusoidal Model

$$y[n] = \sum_{r=1}^R A_r[n] \cos(2\pi f_r[n]n)$$

R : No. of sinusoids

$A_r[n]$: instantaneous amplitude

$f_r[n]$: instantaneous frequency (Hz)

Sinusoids as spectral peaks

Sinusoid $\rightarrow$ peak in magnitude spectrum

Frequency resolution: $-(1/2)$ bin

Improvement of frequency resolution by:-

- zero-padding
- spectral interpolation

Spectral Peaks and Window Size

$$\text{If } B_f = B_s f_s / M \quad \text{and} \quad \Delta = |f_{k+1} - f_k|$$

B = main lobe bandwidth of window

f_s = sampling rate in Hz

M = window size

f_k and f_{k+1} = frequency of sinusoids in Hz.

$$M \geq B_s \frac{f_s}{\Delta} = B_s \frac{f_s}{|f_{k+1} - f_k|} \quad \left(\text{to distinguish the peaks of } f_k \text{ and } f_{k+1} \right)$$

$$\text{If } \Delta = \delta_0 \text{ then } B_s \leq \delta_0 \text{ and } M \geq B_s f_s / \delta_0, \text{ or } M \geq B_s P,$$

where P = period in samples.

for eg: Using Hamming window and distinguishing b/w 440 and 490 Hz

$b_s = 4$ (for Hamming window)

$$f_s = 44100 \text{ Hz}$$

$$\therefore M \geq 9 \left(\frac{44100}{490 - 440} \right) = 3528$$

$$f_k = 440 \text{ Hz}, f_{k+1} = 490$$

$\therefore$ Taking window size $(M) \geq 3528$ will be able to distinguish b/w 440 and 490 Hz

Peak Detection

p_r = peak of sinusoid = $mX[k_0]$ when $mX[k_0-1] < mX[k_0] > mX[k_0+1]$

mX = magnitude spectrum

r = peak number

k_0 = peak location

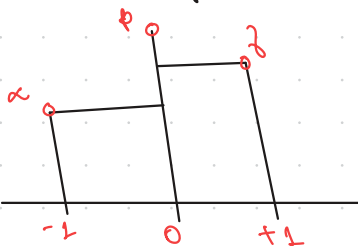
To increase resolution, we use zero padding

We can use parabola interpolation

Parabola, $x[n] = a(n-p)^2 + b$

p : centre of parabola
 a : concavity measure
 b : offset

Peak interpolation



$$x[-1] = \alpha = mX[k_p - 1]$$

$$x[0] = \beta = mX[k_p]$$

$$x[1] = \gamma = mX[k_p + 1]$$

center of parabola : $\hat{k}_p = k_p + \frac{1}{2} \frac{\alpha - \delta}{(\alpha - 2\beta + \delta)}$

amplitude : $\hat{a} = \beta - \frac{1}{4} (\hat{k}_p - k_p)(\alpha - \delta)$

Sinusoidal parameters from peaks
calculated peak location

True peak location $\hat{k}_p = k_p + \frac{0.5 * (|x[k_p+1]| - |x[k_p+2]|)}{|x[k_p-1]| - 2 * |x[k_p]| + |x[k_p+1]|}$

$s_p = \frac{s_s * \hat{k}_p}{N}$

$A_p = |x[k_p]| - 0.25 * (|x[k_p-1]| - |x[k_p+1]|) * (\hat{k}_p - k_p)$

$ph_p = \angle \times [\hat{k}_p]$

Sinusoids in spectrogram

Sinusoid :- stable peak track in spectrogram

Stability identified by

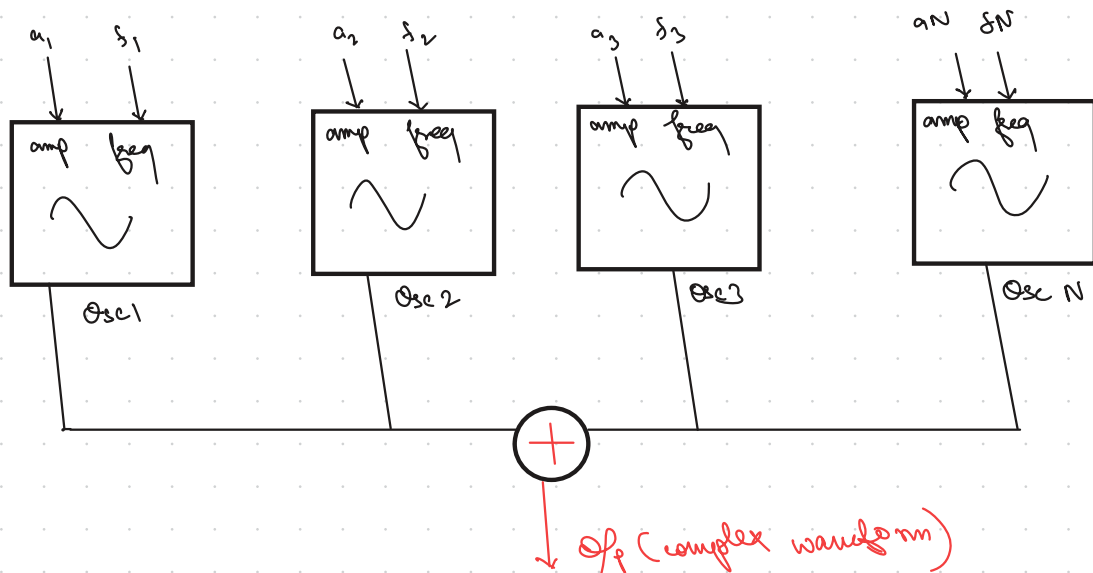
- frequency and amplitude in successive frames
- phase derivative in time/freq

Condition for a peak s_p of frame l to be a part of track t :

$s_p[l] = s_t[l]$ if $(|s_p[l] - s_t[l-1]| < \text{threshold})$
and if exists $s_t[l-2], s_t[l-3], \dots, s_t[l-k]$

⊕ New peak frequency - frequency from prev track < threshold.

Summational (additive) synthesis



To generate a single sine wave

$$y[n] = A_s[n] \cos(2\pi f_s[n]n + \phi_s)$$

$A_s[n]$: instantaneous amplitude $f_s[n]$: instantaneous frequency

ϕ_s : initial phase

Computationally more expensive

Sinusoidal synthesis: discrete frequency

$$y[n] = \text{IDFT}(m_Y[k] * e^{j * p_Y[k]})$$

$m_Y[k] = A_0$ for $k = k_0$ and 0 for $k \neq k_0$; $p_Y[k] = \phi_0$ for $k = k_0$ and 0 for $k \neq k_0$

Sinusoidal synthesis: any frequency

$$y[n] = \text{IDFT} \left(A_0 * m_w[k-k_0] * e^{j * (p_w[k-k_0] + \phi_0)} \right)$$

$m_w[k]$, $p_w[k]$, magnitude and phase spectrum of window

We can do sinusoidal synthesis using only the main lobe of the window.

Additive synthesis

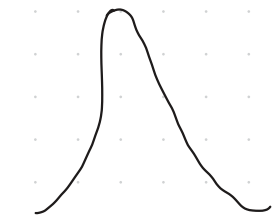
$$y[n] = \text{IDFT} \left(\sum_{r=0}^R A_r * m_w[k-k_r] * e^{j * (p_w[k-k_r] + \phi_r)} \right)$$

$m_w[k]$, $p_w[k]$ magnitude and phase spectrum of the window main lobe

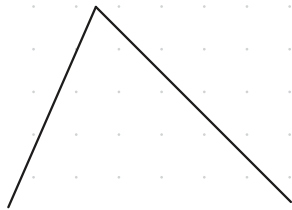
Synthesis Windows

To reduce overlap size,

$$y_w = y * \frac{\text{triangular}}{\text{Blackman Harris}}$$



Blackman-Harris

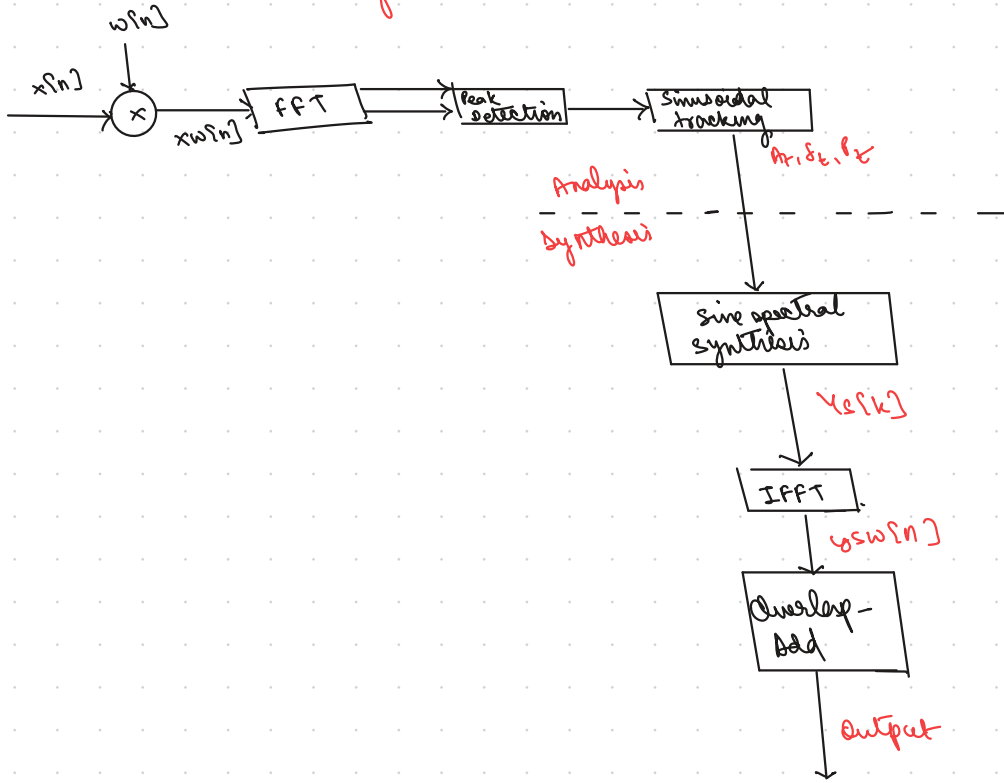


Triangular



triangular / Blackman Harris.

Sinusoidal model system



Harmonic Model

$$y_h[n] = \sum_{r=1}^R A_r[n] \cos(2\pi r f_0[n] n)$$

$$r \in \mathbb{Z}$$

R : No. of harmonic components
 $A_r[n]$: instantaneous amplitude
 $f_0[n]$: fundamental frequency

Spectral View

$$Y_{h_r}[k] = \sum_{r=1}^R A_{(r,L)} w[k-r \hat{f}_{(r,L)}]$$

w : spectrum of analysis window

R : No. of harmonics

A : amplitude of harmonic

$\hat{f}$: normalized fundamental frequency

L : frame number

r : harmonic number

Harmonic Detection

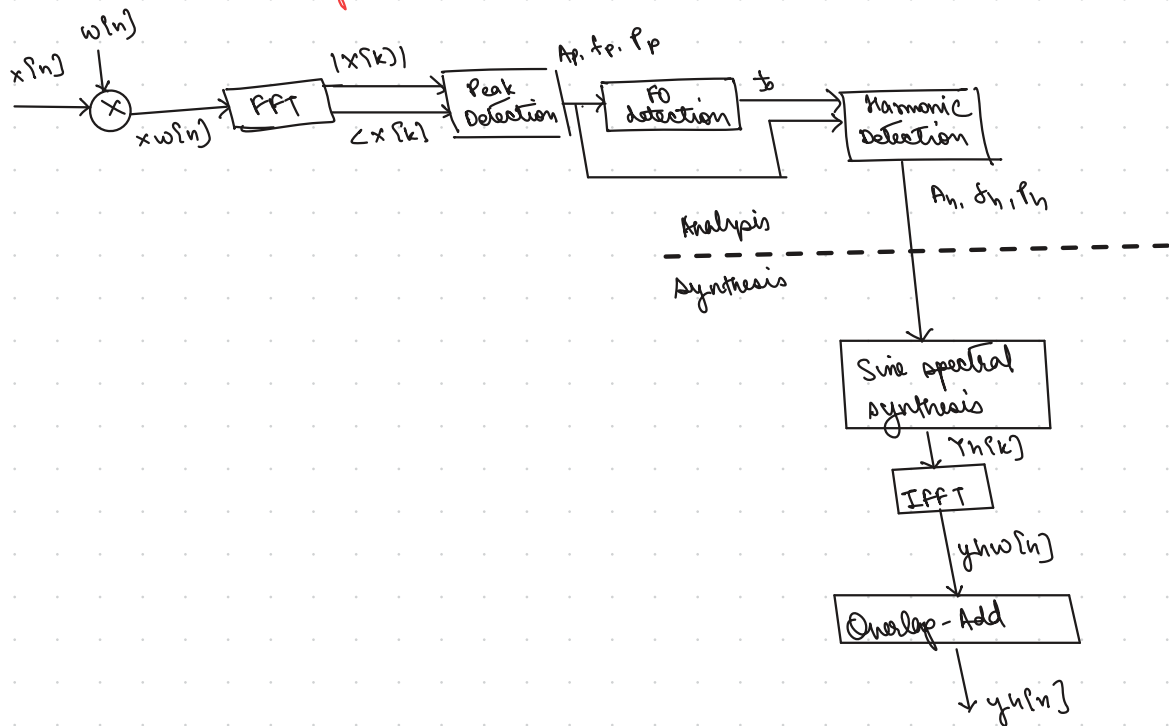
Harmonic $\rightarrow$ stable spectral peak whose frequency is close to a multiple of the fundamental frequency.

Condition for a peak f_p to be a harmonic f_h :

$$f_p[l] = f_h[l] \text{ if } (|f_p[l] - (h * f_0[l])| < \text{threshold})$$

and exists $f_h[l-2], f_h[l-3], \dots, f_h[l-L]$

Harmonic Model System



Fundamental Frequency Detection

- FO detection in time domain
 - monophonic signals
- FO detection in frequency domain
 - monophonic and polyphonic signals.

FO detection in time domain

Auto correlation function (with tapering)

$$r_x[l] = \sum_{n=0}^{N-l-1} x[n] x[n+l] \quad l=0, 1, 2, \dots, N-1$$

where $l = \text{lag}$

YIN Algorithm (in time domain)

$$d[l] = \sum_{n=0}^{N-l-1} (x[n] - x[n+l])^2 \quad l=0, 1, 2, 3, \dots, N-1$$

FO detection in frequency domain

FO in the spectrum :- Can be defined as the common divisor of the harmonic series that explains the spectral peaks.

Pattern Matching (Two way mismatch algorithm)

$$\begin{aligned} \text{Err}_{p \rightarrow m} &= \sum_{n=1}^N E_{\omega}(\Delta \delta_n, f_n, a_n, A_{\max}) \\ (\text{predicted} \rightarrow \text{measured}) \\ &= \sum_{n=1}^N \Delta \delta_n \cdot (\delta_n)^p + \left(\frac{a_n}{A_{\max}} \right) \times [q \delta_n (f_n)^p - r] \end{aligned}$$

$\Delta \delta_n$: difference b/w predicted and closest measured peaks

f_n, a_n : frequency and magnitude of predicted peaks

$A_{\max}$: max^m peak magnitude

Frequency (Hz)	predicted	measured
700		
600		
550		
450		
350		
250		
200		
100		

predicted measured

Algorithm :-

We take every predicted peak value and look at the closest measured peak value

We then find the frequency distance and scale it to an amplitude

$$\begin{aligned} \text{Err}_{m \rightarrow p} &= \sum_{k=1}^K E_w(\Delta s_k, f_k, a_k, A_{\max}) \\ &= \sum_{k=1}^K \Delta s_k (f_k)^p + \left(\frac{a_k}{A_{\max}} \right) \times [q \Delta s_k \cdot (f_k)^p - r] \end{aligned}$$

measured $\rightarrow$ predicted

Δs_k :- diff b/w predicted and its closest measured peaks

f_k, a_k :- frequency magnitude of predicted peaks

$A_{\max}$:- max^m peak magnitude

Total Error:-

$$E_{\text{err total}} = \text{Err}_{p \rightarrow m} / N + SE_{\text{err}_{m \rightarrow p}} / K$$

Maker and Beauchamp propose :- $p=0.5$, $q=1.4$, $r=0.5$, $s=0.33$

Example :-

	$\text{Err}_{p \rightarrow m}$	$\text{Err}_{m \rightarrow p}$	$E_{\text{err total}}$
50 Hz	132.58	-3.0	7.49
100 Hz	32.0	-3.0	3.93
200 Hz	10.0	30.66	4.2

TWM error calculation from the frequencies: 200, 300, 400, 500, 600, 700, 800

Here at 100 Hz, error is least but not present in the frequencies. Fundamental frequency (F0) may not represent in the list of frequencies

Stochastic Model

Stochastic Model of Signal

$$y[n] = \sum_{k=0}^{N-1} u[n] h[n-k]$$

$u[n]$: white noise

$h[n]$: impulse response of filter after approximating i/p signal $x[n]$

Spectral View

$$Y[k] = |H[k]| |U[k]| e^{j(\angle H[k] + \angle U[k])} = |\tilde{x}[k]| e^{j\angle U[k]}$$

$|\tilde{x}[k]|$: approximation of magnitude spectrum of i/p signal $x[n]$

$\angle U[k]$: spectral phase of noise signal.

l : frame number

→ Linear Predictive Coding

LPC approximation

$$\hat{x}[n] = \sum_{k=1}^K a_k x[n-k]$$

$$\text{Error} = \sum_{n=-\infty}^{\infty} \left(x[n] - \sum_{k=1}^K a_k x[n-k] \right)^2$$

Envelope Approximation

$$\tilde{a}[k] = \text{IDFT}(\text{LP}(\text{DFT}(a[k])))$$

$$b[k] = \text{IDFT}(\text{ZP}(\text{DFT}(\tilde{a}[k])))$$

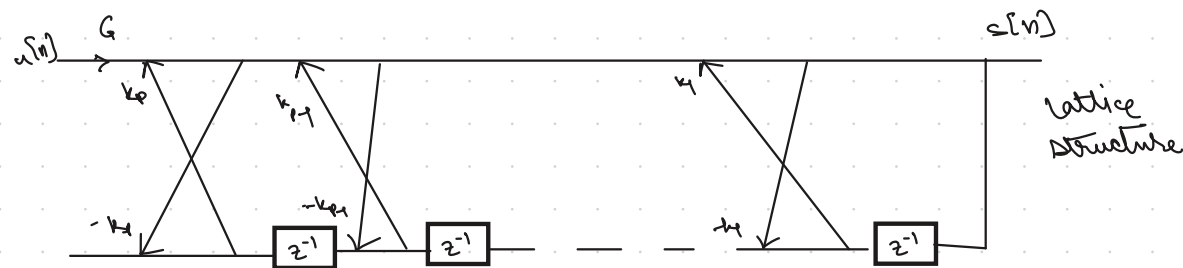
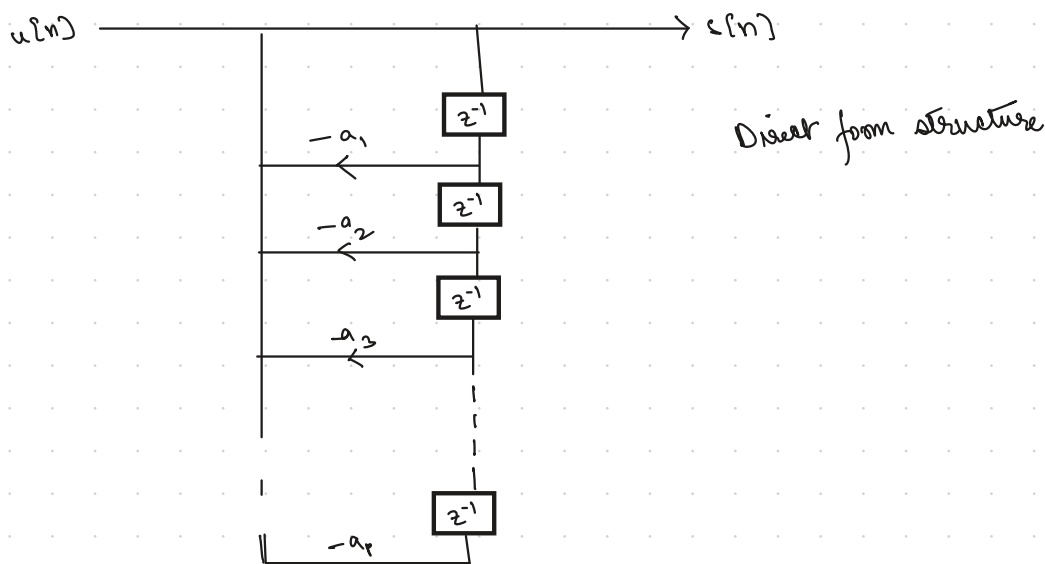
LP: Low-Pass Filter

ZP: zero-padding

$\tilde{a}[k]$ has less no. of samples so zero padding is done so that the no. of samples of $a[k]$ and $b[k]$ are same

Stochastic synthesis using LPC

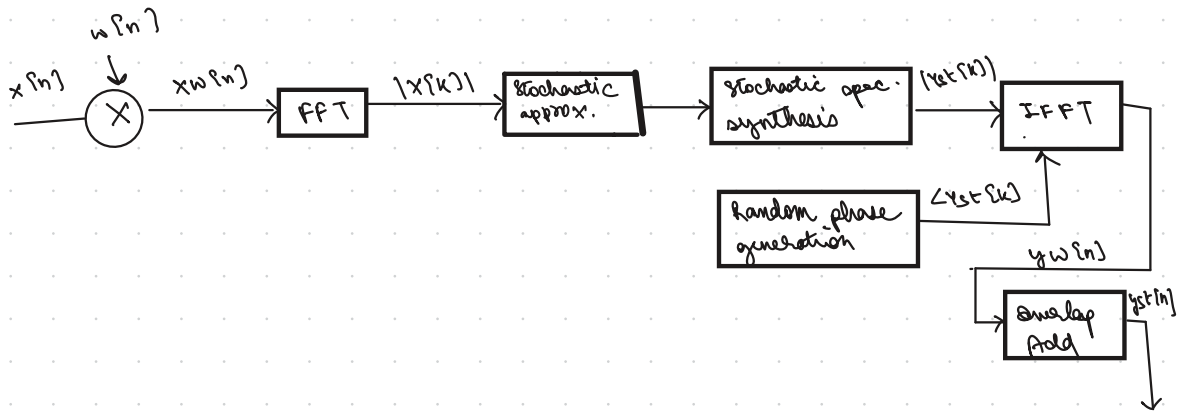
$$y_{\text{st}}[n] = \sum_{k=1}^K a_k u[n-k] \quad a_k: \text{filter coefficients}; u[n]: \text{white noise}$$



Stochastic synthesis using envelopes

$$y_{\text{st}}[n] = \text{IDFT}(\tilde{X}[k] | e^{j\phi[k]})$$

Stochastic Model System



Sinusoidal Plus Harmonic Modelling

$$y[n] = \sum_{r=1}^R A_r[n] \cos(2\pi f_r[n]n) + x_r[n] = y_s[n] + x_r[n]$$

- R : No. of sinusoidal components
- $A_r[n]$: instantaneous amplitude
- $f_r[n]$: instantaneous frequency
- $x_r[n] = x[n] - y_s[n]$: residual component
- $y_s[n]$: sinusoidal component

Spectral View

$$Y_e[k] = \sum_{r=1}^R A_{(r,e)} w[k - \hat{f}_{(r,e)}] + x_{r_e}[k] = y_{s_e}[k] + x_{r_e}[k]$$

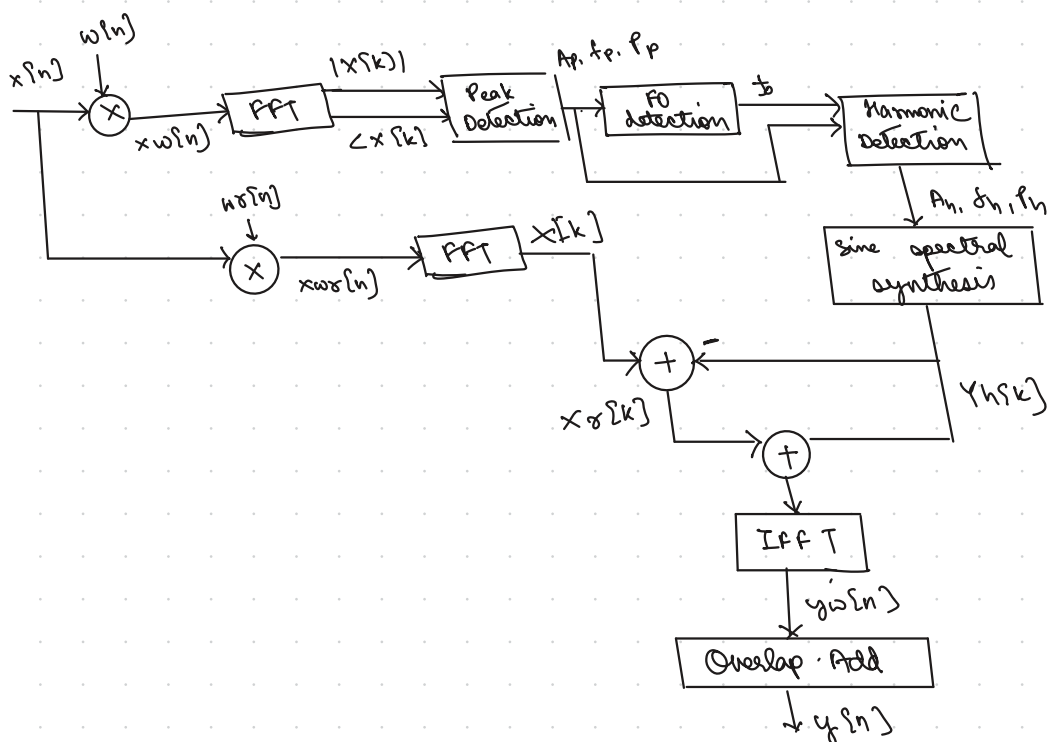
- $W[k]$: Spectrum of analysis window
- R_e : No. of sinusoidal components
- $A_{(r,e)}$: Amplitude of sinusoid
- $\hat{f}_{(r,e)}$: Normalized frequency of sinusoid

$x_{r_l}[k] = x_l[k] - y_{s_l}[k]$: residual component spectrum

$y_{s_l}[k]$: sinusoidal component spectrum

l : frame Number

Hph model system



Sinusoidal + Stochastic Model

$$y[n] = \sum_{s=1}^K A_s[n] \cos(2\pi f_s[n]n) + y_{st}[n] = y_{s}[n] + y_{st}[n]$$

K : No. of sinusoidal components
 $A_s[n]$: instantaneous amplitude
 $f_s[n]$: instantaneous frequency (Hz)

$y_{st}[n]$: stochastic component.
 $y_s[n]$: sinusoidal component.

where

$$y_{st_l}[n] = \sum_{k=0}^{N-1} u[n] \eta[n-k]$$

$u[n]$: white noise

$\eta[n]$: impulse response of residual approximation

l : frame number.

Spectral View

$$Y_l[k] = \sum_{r=1}^R A_{(r,l)} W[k - \hat{f}_{(r,l)}] + x_{st_l}[k]$$

$W[k]$: spectrum of analysis window

R : number of sinusoidal components

$A_{(r,l)}$: amplitude of sinusoid

$\hat{f}_{(r,l)}$: normalized frequency of sinusoid

l : frame number

where

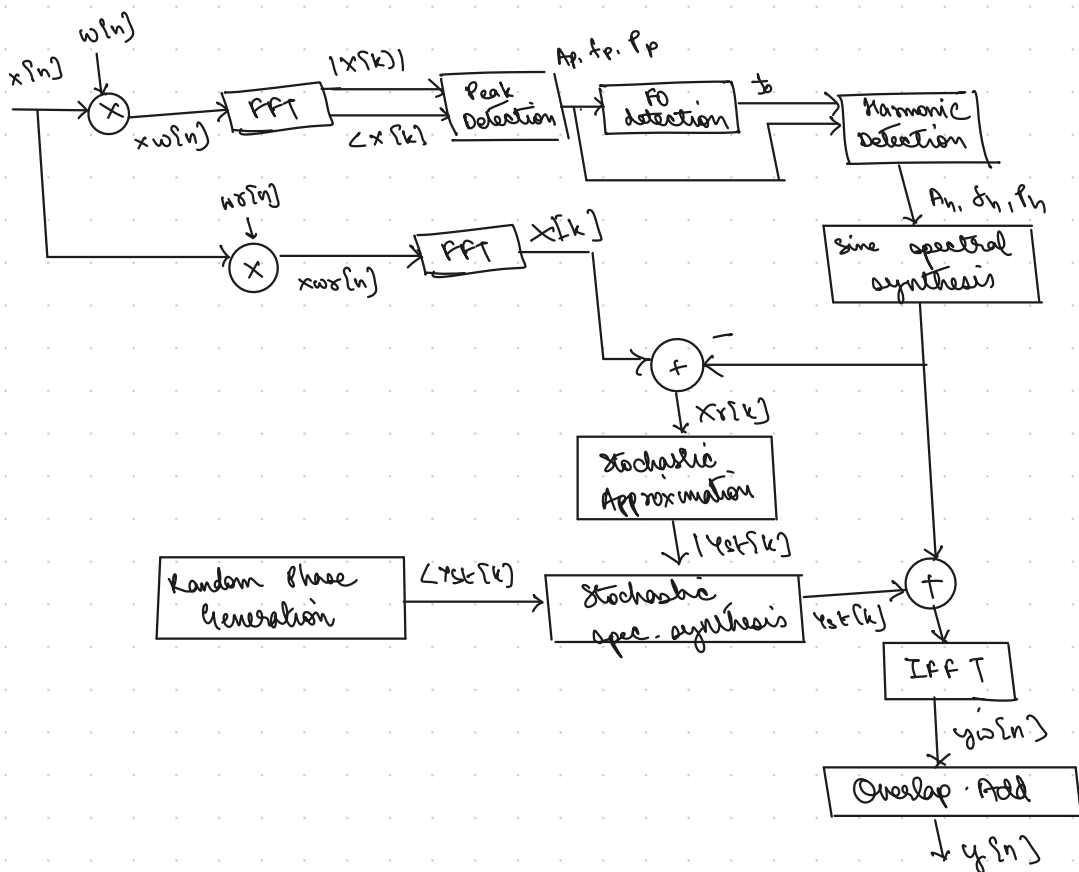
$$x_{st_l}[k] = |\tilde{x}_{\sigma_l}[k]| e^{j\phi[k]}$$

$|\tilde{x}_{\sigma_l}[k]|$: magnitude approximation of residual

$\phi[k]$: spectral phases of noise

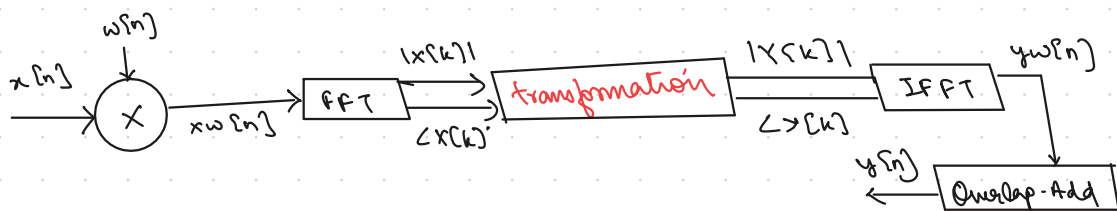
l : frame number

H_pS model system

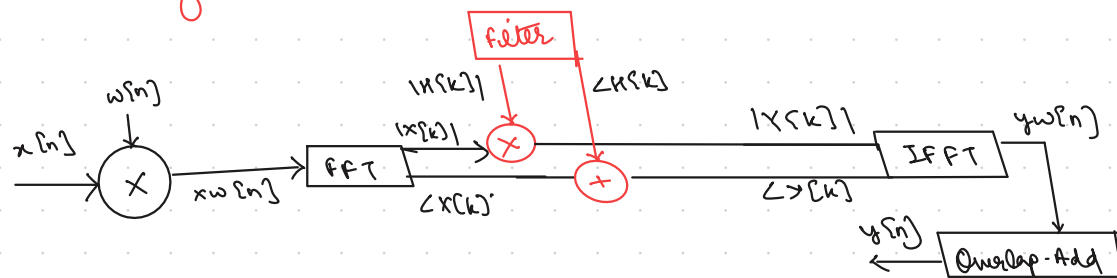


Sound Xformation

STFT Model

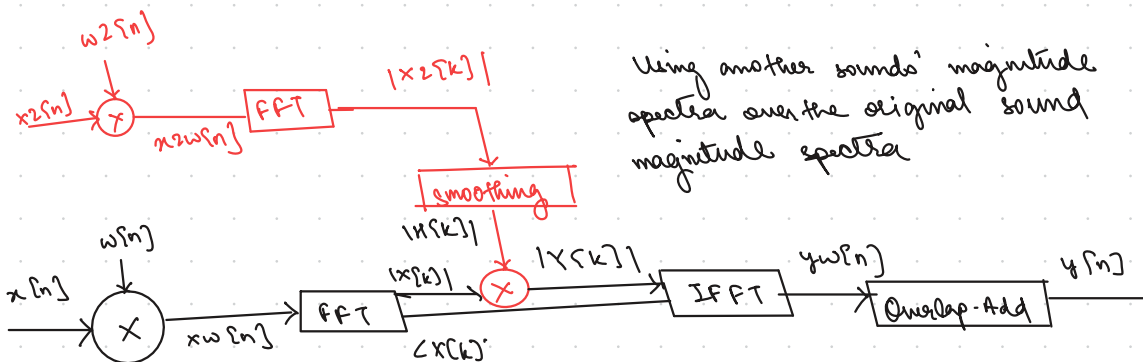


Filtering with STFT



$$Y[k] = |H[k]| |X[k]| e^{j(\angle H[k] + \angle x[k])}$$

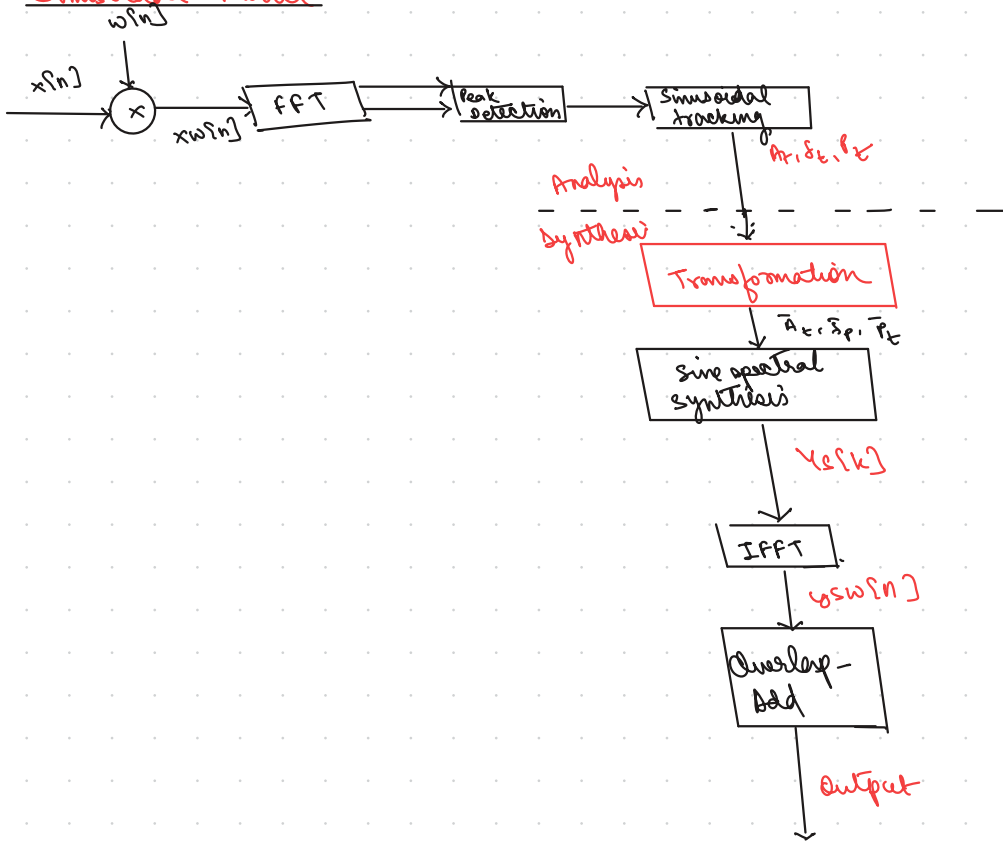
Morphing with STFT



Using another sound's magnitude spectrum over the original sound magnitude spectrum

$$Y[k] = |x_2[k]| |x_1[k]| e^{j\angle x_1[k]}$$

Sinusoidal Model



Scaling frequency, amplitude, Time

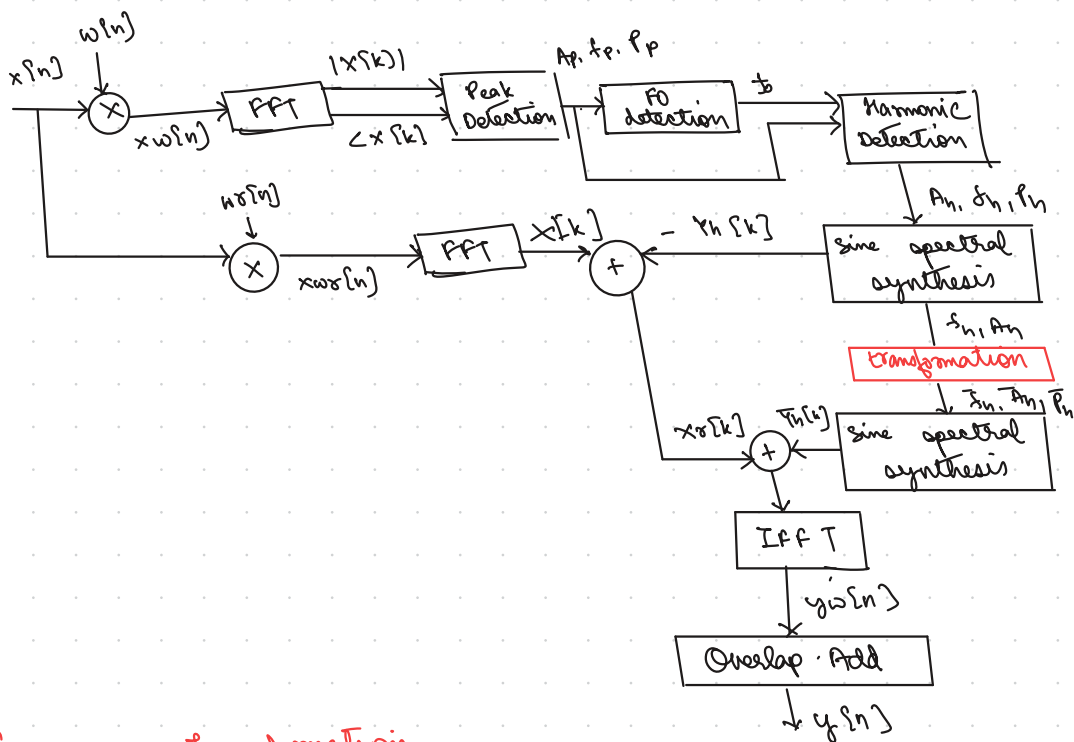
$$\bar{s}_t[q] = s_{s_t}[l] s_t[s_{s_t}[l]l]$$

$$\bar{A}_t[q] = sA_t[l] + A_t[s_{s_t}[l]l]$$

$$\bar{\phi}_t[q] = \phi_t[q-1] + s_{s_t}[q]$$

q : o/p frame index; l : input frame index; t : sinusoidal track index
 f : i/p frequency in Hz; A : i/p amplitude in dB
 s_t : scaling freq.; $\bar{A}$: o/p amplitude; $\bar{\phi}$: o/p phase

Harmonic plus residual model



Frequency Transformation

Frequency Transposition

$$\bar{f}_h[l] = sf[l] f_h[l]$$

Frequency Shifting

$$\bar{f}_h[l] = sf[l] + f_h[l]$$

Frequency stretching

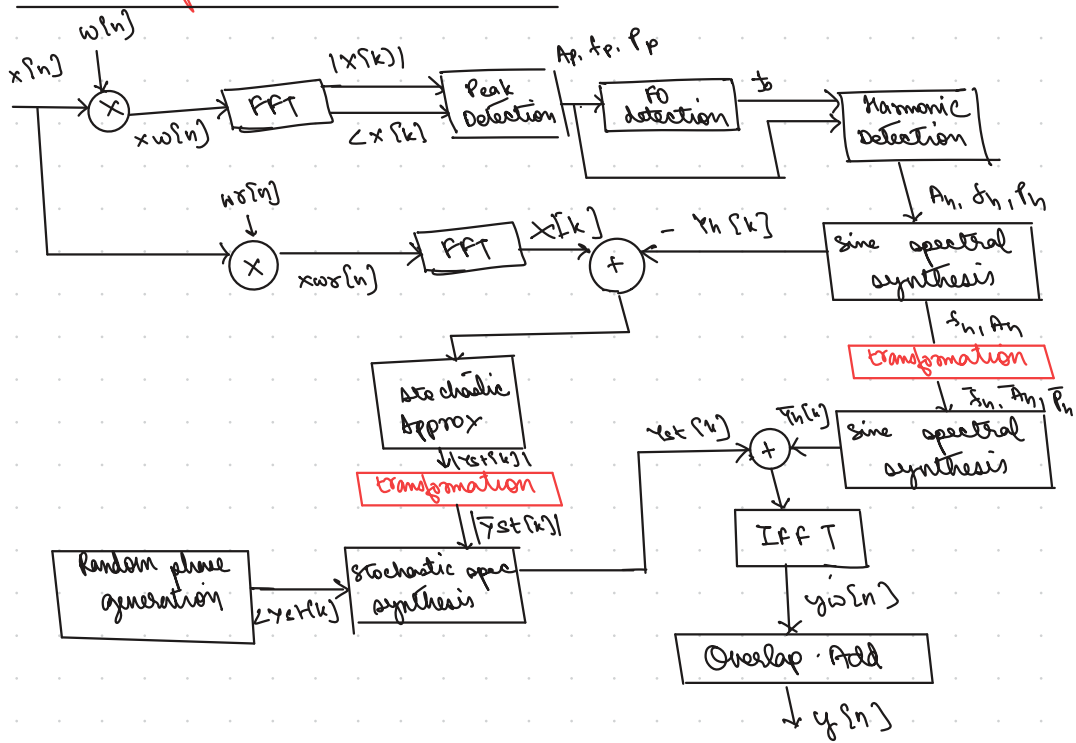
$$\bar{f}_h[l] = (f_h[l] / h) \times h^{sf[l]}$$

f_h : input frequency of harmonic h

$\bar{f}_h$: of p frequency of harmonic h

sf : scaling frequency

Harmonic plus stochastic model



Scaling amplitude, frequency and time

$$\hat{f}_h[q] = s f_h[l] \delta_k [st_h[l]l]$$

$$\hat{A}_h[q] = s A_h[l] + A_k [st_h[l]l]$$

$$\hat{\varphi}_h[q] = \varphi_h[q+1] + \delta_h[q]$$

$$s \hat{E}_k[q] = s s t_k[l] s t_k [st_k[l]l]$$

q : o/p frame ; l : i/p frame ; h : harmonic

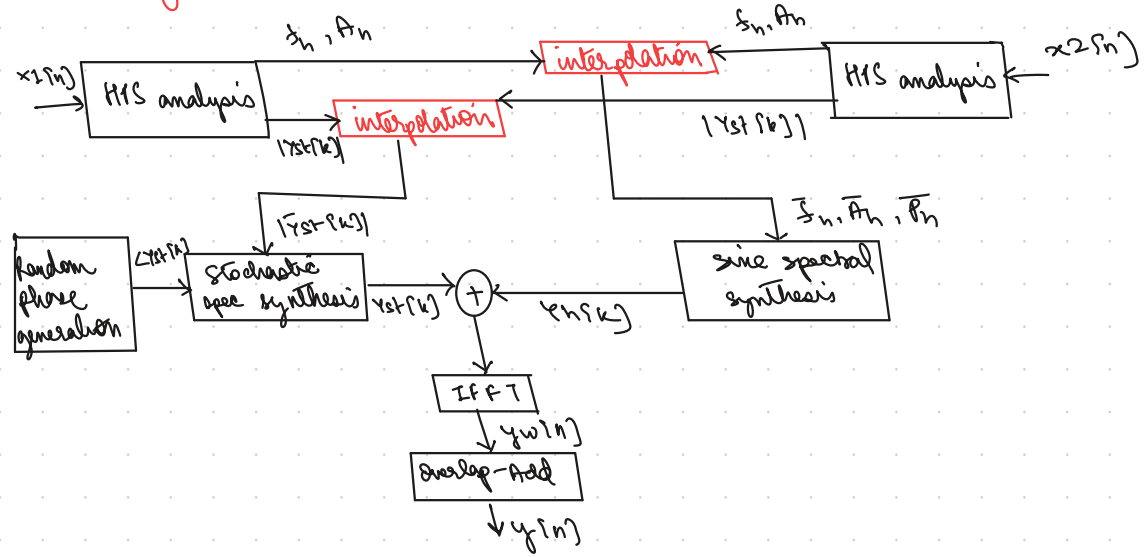
s : i/p frequency ; A : i/p amplitude ; st : i/p stochastic envelope

$s f$: scaling frequency ; $s A$: scaling amplitude ; $s t$: scaling time ;
 $s s t$: scaling stochastic.

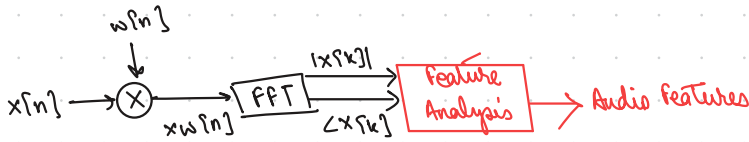
$\bar{f}$: of frequency ; $\bar{A}$ = of amplitude ; $\bar{\varphi}$ = of phase

st : of stochastic.

Morphing



Audio features



Single frame Spectral Features

- Energy, RMS, loudness
- Spectral centroid
- Mel-frequency cepstral coefficients (MFCC)
- Pitch salience
- Chroma (harmonic pitch class profile, HPCP)

Energy, RMS and loudness

Energy,

$$\text{energy}_x = \sum_{k=0}^{N-1} |x_x[k]|^2$$

Root Mean Square

$$\text{RMS}_x = \sqrt{\frac{1}{N} \sum_{k=0}^{N-1} |x_x[k]|^2}$$

Steven's Power Law

$$\text{loudness}_x = \left(\sum_{k=0}^{N-1} |x_x[k]|^2 \right)^{0.67}$$

Spectral Centroid

$$\text{Centroid}_x = \frac{\sum_{k=0}^{N/2} k |x_x[k]|}{\sum_{k=0}^{N/2} |x_x[k]|}$$

Mel frequency cepstral coefficients

$$\text{mfcc}_x = \text{DCT} \left(\log_{10} \left(\sum_{k=0}^{N/2} |x_x[k]| H_i[k] \right) \right)$$

where

$|x_x[k]|$ is the positive magnitude spectrum

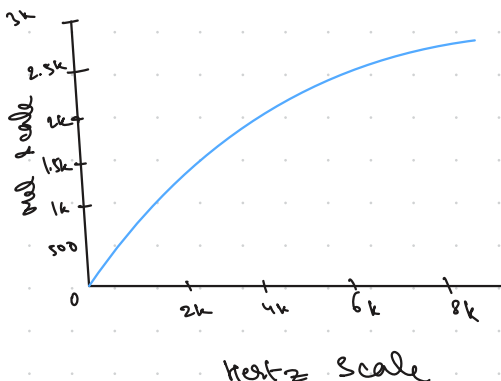
$H_i[k]$ is the mel scale filter bank for each filter i

$$\text{DCT}[m] \text{ (discrete cosine transform)} = \sum_{n=0}^{N-1} f[n] \cos \left(\frac{\pi}{2} (n+1/2) m \right)$$



MFCC : Mel Scale

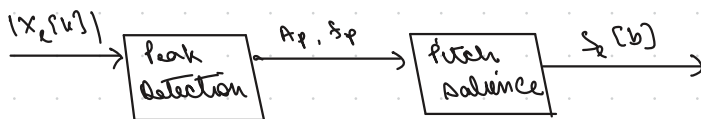
$$\text{mel} = 2595 \log_{10} \left(1 + \frac{f}{700} \right)$$



zeroth coefficient describes the loudness of the sound

1st coefficient gives the general picture and as coefficient no. increases, more details are shown

Pitch Salience



$$S[b] = \sum_{n=1}^N \sum_{p=1}^P e(A_p) g(b, n, \delta_p) (A_p)^{\beta}$$

where

$S[b]$ = salience at bin frequency b (b expressed in cent scale)

$e()$ = magnitude threshold function

$g()$ = weighting function applied to peak p

β = magnitude compression value

It is a mathematical scoring system in DSP used to determine the fundamental pitch of a complex sound. Instead of simply picking the lowest frequency in the spectrum, the algorithm tests every possible 'musical note' and calculates a 'confidence score' based on how well the audio supports that guess.

It adds up the points gathered from all the harmonics for every candidate pitch. The frequency that achieves the highest total score is called the fundamental note.

Example

The Setup

You plug in, kick on a high-gain distortion pedal, and heavily strike the **open A string (110 Hz)**.

Because of the string physics and the distortion, the STFT analysis doesn't just output a single clean 110 Hz wave. The algorithm detects a whole mess of peaks (p):

- **50 Hz:** Electrical ground hum from your amp.
- **110 Hz:** The fundamental note (A2).
- **220 Hz, 330 Hz, 440 Hz:** The natural, screaming harmonics of the string.
- **2500 Hz:** The harsh, non-musical scratch of your guitar pick hitting the strings.

Test 1: The algorithm guesses you played an A (110 Hz)

The algorithm starts its nested loop.

Loop 1 (The Harmonics, h): Since the guessed pitch is 110 Hz, it knows the ideal harmonics should be exactly at 110, 220, 330, and 440 Hz.

Loop 2 (The Peaks, p): For each ideal harmonic, it scans through the actual peaks it found in the audio to see if there is a match.

Here is how the three parts of the equation react:

- 1 $e(A_p)$ — **The Noise Gate:** The algorithm looks at the 50 Hz amp hum and the 2500 Hz pick scrape. Because their volumes (A_p) are below a certain threshold, $e()$ simply equals 0. This multiplies their score by zero, instantly dropping them from the equation so noise doesn't confuse the pitch detector.
- 2 $g(b, h, f_p)$ — **The Matchmaker:** The algorithm expects a 2nd harmonic at 220 Hz. It looks at your real audio peaks and says, "Bingo! There is a loud peak sitting exactly at 220 Hz." Because the frequencies match perfectly, $g()$ outputs a high multiplier (e.g., 1.0).
- 3 $(A_p)^\beta$ — **The Compressor:** Let's say the fundamental 110 Hz peak is overwhelmingly loud. If we just added the raw volumes together, that single peak might hijack the whole score. The β (a decimal like 0.5) mathematically compresses the loudness ↓ like a studio compressor plugin. This ensures the quiet but crucial higher harmonics (like 330 Hz and 440 Hz) carry enough weight to "prove" this is a

Result for A (110 Hz): Every expected harmonic finds a matching, sufficiently loud peak in the real audio. The score $S[b]$ skyrockets.

Test 2: The algorithm guesses you played a D (146 Hz)

Now the loop runs again, testing a wrong note.

Loop 1 (The Harmonics, h): If the note is D, the ideal harmonics should be at 146 Hz, 292 Hz, and 438 Hz.

Loop 2 (The Peaks, p): The algorithm scans your real audio peaks (which are still sitting at 110, 220, and 330 Hz).

- **The Matchmaker $g()$:** The algorithm takes the expected 146 Hz and scans the real peaks. 110 is too low. 220 is too high. There is no match. $g()$ outputs a 0.
- It tries the next harmonic (292 Hz). Again, no matching peak. $g()$ outputs a 0.

Result for D (146 Hz): Because none of the expected harmonic frequencies align with the actual audio data, the entire equation multiplies out to zero. The score $S[b]$ is 0.

The Conclusion

Once the algorithm finishes looping through all the candidate pitches (all the frets), it looks at the final scoreboard. The pitch bin for A has a massive score, while the others have near zero. The algorithm outputs 110 Hz as the ~~fundamental pitch~~.

Chroma (Harmonic Pitch Class Profile)

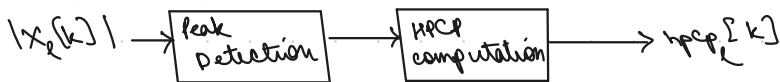
$$hpcp[k] = \sum_{p=1}^P w(k, f_p) A_p^2$$

where

A_p = amplitude of spectral peak p

$w(k, f_p)$ = weight of the peak frequency f_p for bin k

k = spectral bin locations of the chosen HPCF frequencies



While pitch salience identifies a single fundamental note chroma (HPCF) is used to analyze the harmonic content of sound. It is the core algorithm used by computers to identify the chord being played or detect the musical key of a song.

Octave Equivalence: HPCP takes a massive, complex frequency spectrum and collapses it down to 12 distinct bins representing the 12 notes of the Western musical scale (C, C#, D, D#, E, F, F#, G, A, A#, B) to completely ignoring which octave the notes are in.

How the Algorithm Works:

The algorithm loops through every detected spectral peak in an audio frame and routes its energy into one of those 12 bins using two main steps:

- 1 **Energy Calculation (A_p^2):** The algorithm squares the amplitude of each peak. This mathematically suppresses quiet background noise and exponentially highlights the loud, musically important frequencies.
- 2 **Sorting (Weighting):** A weighting function analyzes the exact frequency of the peak, determines which musical note it belongs to, and drops its squared energy into that specific pitch class bin.

The Result:

The final output is a simple 12-bar profile showing the total accumulated energy for each pitch class. For example, if a dense, distorted E minor chord is played across all six guitar strings, the resulting HPCP will show massive, towering spikes in only the E, G, and B bins, allowing the software to instantly recognize the chord geometry.

Multiple frames spectral features

Event segmentation

Spectral flux (used in segmentation)

$$SF_L = \sum_{k=0}^{N/2} H(|x_L[k]| - |x_{L-1}[k]|)$$

where $H(x) = \frac{x + |x|}{2}$

Onset Detection (based on high frequency content)

$$\text{Onset Detection function} = HFC_L - HFC_{(L-1)}$$

where $HFC_L = \sum_{k=1}^{N/2} |x_L[k]| k^2$

Predominant Pitch



Sound and Music Description

lower to higher level

cognitive	emotion, music style, semantic concepts			
formal	melody, key, tonality	rhythmic patterns, tempo, meter	instrument, voice	articulation
perceptual	successive and simultaneous intervals	time (beat)	timbre (spectral envelope)	dynamics
sensorial	pitch	time	timbre	loudness
physical	frequency	duration (onset)	spectrum (centroid)	intensity

Similarity b/w sounds

Euclidean distance

$$d(p, q) = \sqrt{\sum_{i=1}^n (q_i - p_i)^2}$$

Clustering of sounds

K Means

$$\arg \min_s \sum_{i=1}^k \sum_{x \in S_i} \|x - \mu_i\|^2$$

where $(x_1, x_2, x_3, \dots, x_n)$ are the set of observations, μ_i is the mean of points in set S_i

k nearest neighbour classifier (KNN)

